

A mini-course on concentration inequalities

Introduction

Toss a fair coin a million times. The individual outcomes are unpredictable, yet the total number of heads is almost certain to land very near half a million. Randomness in the small produces near-certainty in the large. This is *concentration*: although each piece is unpredictable, a quantity assembled from many independent random pieces can be almost deterministic.

Concentration inequalities make this precise. The central limit theorem describes fluctuations only in the limit $n \rightarrow \infty$. A concentration inequality instead gives an explicit failure probability valid for every sample size n – exactly what one needs to guarantee a procedure run on finite data.

This self-contained mini-course develops four such inequalities: Hoeffding's inequality, Hoeffding's inequality for sampling without replacement, McDiarmid's inequality, and Bernstein's inequality. All four are proved using the same technique, the moment-generating-function method. The mini-course then applies them to prove a cornerstone of statistical learning theory: a learner that minimizes error on its training sample does, with high probability, almost as well on unseen data as the best hypothesis available to it.

The logical dependencies are as follows.

- Markov's inequality $\rightarrow$ Chernoff's bound $\rightarrow$ the MGF method.
- Hoeffding's lemma $\rightarrow$ Hoeffding's inequality for independent bounded sums.
- Conditional Hoeffding lemma $\rightarrow$ martingale bounded-differences bounds.
- Martingale bounded differences $\rightarrow$ sampling without replacement and McDiarmid.
- Moment expansion plus Chernoff $\rightarrow$ Bernstein.
- Symmetrization plus sampling without replacement plus McDiarmid $\rightarrow$ the i.i.d. growth-function ERM guarantee.

What concentration means

A concentration inequality is a finite-sample statement that a random quantity Z is unlikely to be far from a typical value, usually its mean:

$$\mathbb{P}[|Z - \mathbb{E}[Z]| \geq t] \leq \text{small}(t),$$

where the useful cases have a right-hand side that decays exponentially in the sample size or in t^2 .

To make this concrete, return to the coin. Let $X_1, \dots, X_n \in \{0, 1\}$ be independent with $\mathbb{E}[X_i] = p$, and let $A_n := \frac{1}{n} \sum_{i=1}^n X_i$. Hoeffding's inequality (proved below) gives

$$\mathbb{P}[|A_n - p| \geq \varepsilon] \leq 2 \exp(-2n\varepsilon^2).$$

The right-hand side stays close to 1 while ε is of order $\frac{1}{\sqrt{n}}$ or smaller, and decays exponentially once ε exceeds that scale. Thus deviations much larger than $\frac{1}{\sqrt{n}}$ are exponentially unlikely, and $\frac{1}{\sqrt{n}}$ is the typical scale of $|A_n - p|$. This is the central-limit scale, but here it comes with an explicit bound valid for every n .

The intuition is cancellation plus stability: no single bounded observation can move the average much, and independent fluctuations do not consistently agree in sign. For a general function $f(X_1, \dots, X_n)$, concentration should persist if changing one coordinate changes f only slightly – the bounded-differences idea behind McDiarmid's inequality.

The common MGF technique

For a real random variable Z , define its log moment-generating function (MGF)

$$\psi_{Z(\lambda)} := \log(\mathbb{E}[\exp(\lambda Z)]).$$

For $\lambda > 0$, Markov's inequality gives the Chernoff bound

$$\mathbb{P}[Z \geq t] = \mathbb{P}[\exp(\lambda Z) \geq \exp(\lambda t)] \leq \exp(-\lambda t) \mathbb{E}[\exp(\lambda Z)] = \exp(\psi_{Z(\lambda)} - \lambda t).$$

Since this holds for every $\lambda > 0$, the best bound comes from optimizing over λ . Equivalently,

$$\mathbb{P}[Z \geq t] \leq \exp(-\psi_Z^*(t)), \quad \psi_Z^*(t) := \sup_{\lambda > 0} (\lambda t - \psi_{Z(\lambda)}).$$

Independence makes this powerful for sums: if $Z = \sum_{i=1}^n Z_i$ with the Z_i independent, then $\psi_{Z(\lambda)} = \sum_{i=1}^n \psi_{Z_i}(\lambda)$, so it suffices to control one summand. The key case is a sub-Gaussian MGF bound $\psi_{Z(\lambda)} \leq v \frac{\lambda^2}{2}$ for all λ ; Chernoff with $\lambda = \frac{t}{v}$ then gives

$$\mathbb{P}[Z \geq t] \leq \exp\left(-\frac{t^2}{2v}\right).$$

The same idea works conditionally for martingales. Let D_i be martingale differences for a filtration $\mathcal{F}_i$, so $\mathbb{E}[D_i \mid \mathcal{F}_{i-1}] = 0$. If $\mathbb{E}[\exp(\lambda D_i) \mid \mathcal{F}_{i-1}] \leq \exp\left(\lambda^2 \frac{c_i^2}{8}\right)$ almost surely, then iterating conditional expectations gives

$$\mathbb{E}\left[\exp\left(\lambda \sum_{i=1}^n D_i\right)\right] \leq \exp\left(\frac{\lambda^2}{8} \sum_{i=1}^n c_i^2\right),$$

and Chernoff yields a Hoeffding-style tail. Sampling without replacement and McDiarmid's inequality are both applications of this martingale variant.

Statements and consequences

Hoeffding's inequality. Let $X_1, \dots, X_n$ be independent and suppose $X_i \in [a_i, b_i]$ almost surely. Set

$$S := \sum_{i=1}^n (X_i - \mathbb{E}[X_i]), \quad A := \sum_{i=1}^n (b_i - a_i)^2.$$

Then for every $t > 0$,

$$\mathbb{P}[S \geq t] \leq \exp\left(-2\frac{t^2}{A}\right), \quad \mathbb{P}[|S| \geq t] \leq 2 \exp\left(-2\frac{t^2}{A}\right).$$

It controls sums of independent bounded variables. It uses only the ranges $[a_i, b_i]$, not the variances.

For averages of independent $[0, 1]$ -valued variables, this becomes

$$\mathbb{P}\left[\left|\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i]\right| \geq \varepsilon\right] \leq 2 \exp(-2n\varepsilon^2).$$

Hoeffding without replacement. Let $z_1, \dots, z_N \in [a, b]$ be fixed. Draw $T \subseteq \{1, \dots, N\}$ uniformly among subsets of size n , and set

$$\mu := \frac{1}{N} \sum_{i=1}^N z_i.$$

Then for every $\varepsilon > 0$,

$$\mathbb{P}_T \left[\left| \frac{1}{n} \sum_{i \in T} z_i - \mu \right| \geq \varepsilon \right] \leq 2 \exp \left(-\frac{2n\varepsilon^2}{(b-a)^2} \right).$$

This controls a sample mean from a finite population sampled without replacement. The draws are dependent but less variable than independent draws with replacement, so the rate matches the independent Hoeffding bound; this form is all the ERM proof needs.

McDiarmid's inequality. Let $X_1, \dots, X_n$ be independent random variables taking values in a set $\mathcal{X}$. Suppose $f : \mathcal{X}^n \rightarrow \mathbb{R}$ satisfies the bounded-differences condition

$$|f(x_1, \dots, x_n) - f(x_1, \dots, x_{i-1}, x_{i'}, x_{i+1}, \dots, x_n)| \leq c_i$$

for every coordinate i and all inputs. Then, for $Z := f(X_1, \dots, X_n)$ and every $t > 0$,

$$\mathbb{P}[Z - \mathbb{E}[Z] \geq t] \leq \exp \left(-\frac{2t^2}{\sum_{i=1}^n c_i^2} \right),$$

and the same bound holds for $\mathbb{P}[Z - \mathbb{E}[Z] \leq -t]$. This controls well-behaved functions of many independent variables, not just sums.

Bernstein's inequality. Let $X_1, \dots, X_n$ be independent, mean-zero random variables with $|X_i| \leq b$ almost surely. Let

$$\sigma^2 := \sum_{i=1}^n \text{Var}(X_i).$$

Then for every $t > 0$,

$$\mathbb{P} \left[\sum_{i=1}^n X_i \geq t \right] \leq \exp \left(-\frac{t^2}{2(\sigma^2 + bt/3)} \right).$$

Applying the same bound to $-X_i$ gives the two-sided version with an additional factor of 2.

Bernstein is stronger than Hoeffding when the variance is much smaller than the crude range bound: for $t \ll \frac{\sigma^2}{b}$ the tail is Gaussian, $\exp(-\frac{t^2}{2\sigma^2})$, while for large t boundedness takes over and it is exponential in $\frac{t}{b}$. Hoeffding sees only the worst-case range; Bernstein also sees the variance.

Comparison. The table below summarizes the four inequalities. Hoeffding and Bernstein both control a sum of independent terms and differ only in what they measure: Hoeffding charges every term its full range $(b_i - a_i)$, whereas Bernstein replaces the range by the variance and pays the range only through the lower-order term $b \frac{t}{3}$ — a strict improvement whenever the variance is small. Sampling without replacement controls the same kind of sample mean as Hoeffding but with *dependent* draws; the dependence only helps, so the bound is unchanged. McDiarmid keeps Hoeffding's exponent but replaces “sum” by an arbitrary function with bounded coordinatewise differences, which is what makes it applicable beyond sums.

Inequality	Quantity controlled	Key assumptions	Tail bound
Hoeffding	sum S of independent bounded variables	X_i independent, $X_i \in [a_i, b_i]$	$\exp\left(-2\frac{t^2}{A}\right)$, $A = \sum_i (b_i - a_i)^2$
Hoeffding, without replacement	mean of a size- n sample from a finite population	population values in $[a, b]$, sample drawn without replacement	$2 \exp(-2n\varepsilon^2/(b-a)^2)$
McDiarmid	an arbitrary function $f(X_1, \dots, X_n)$	X_i independent, f has bounded differences c_i	$\exp\left(-2t^2 / \sum_i c_i^2\right)$
Bernstein	sum S of independent bounded variables	X_i independent, mean-zero, $ X_i \leq b$	$\exp(-t^2/(2(\sigma^2 + bt/3)))$

Proofs

We first isolate the lemma that powers the sub-Gaussian bounds.

Hoeffding's lemma. If Y is centered and $Y \in [a, b]$ almost surely, then for every $\lambda \in \mathbb{R}$,

$$\log(\mathbb{E}[\exp(\lambda Y)]) \leq \lambda^2 \frac{(b-a)^2}{8}.$$

Proof. Let $\psi(\lambda) = \log(\mathbb{E}[\exp(\lambda Y)])$. Differentiating under the expectation,

$$\psi'(\lambda) = \frac{\mathbb{E}[Y \exp(\lambda Y)]}{\mathbb{E}[\exp(\lambda Y)]}, \quad \psi''(\lambda) = \frac{\mathbb{E}[Y^2 \exp(\lambda Y)]}{\mathbb{E}[\exp(\lambda Y)]} - \psi'(\lambda)^2.$$

These are exactly the mean and variance of Y under the tilted law $d\frac{Q_\lambda}{d}\mathbb{P} = \frac{\exp(\lambda Y)}{\mathbb{E}[\exp(\lambda Y)]}$. Since Q_λ is supported in $[a, b]$, any random variable with this law lies in $[a, b]$, so by the well-known bound on the variance of a $[a, b]$ -valued variable,

$$\psi''(\lambda) = \text{Var}_{Q_\lambda}(Y) \leq \frac{(b-a)^2}{4}.$$

The variance bound follows because $\mathbb{E}[(Y-a)(b-Y)] \geq 0$, which expands to $\text{Var}(Y) \leq (b - \mathbb{E}[Y])(\mathbb{E}[Y] - a) \leq \frac{(b-a)^2}{4}$. Centeredness gives $\psi(0) = 0$ and $\psi'(0) = \mathbb{E}[Y] = 0$, so Taylor's theorem with integral remainder yields

$$\psi(\lambda) = \int_0^\lambda (\lambda - s) \psi''(s) ds \leq \frac{(b-a)^2}{4} \int_0^\lambda (\lambda - s) ds = \lambda^2 \frac{(b-a)^2}{8}.$$

□

Proof of Hoeffding's inequality. Let $S = \sum_i (X_i - \mathbb{E}[X_i])$. By independence and Hoeffding's lemma,

$$\log(\mathbb{E}[\exp(\lambda S)]) = \sum_{i=1}^n \log(\mathbb{E}[\exp(\lambda(X_i - \mathbb{E}[X_i]))]) \leq \frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2 = \lambda^2 \frac{A}{8}.$$

Chernoff gives, for every $\lambda > 0$,

$$\mathbb{P}[S \geq t] \leq \exp\left(-\lambda t + \lambda^2 \frac{A}{8}\right).$$

The right-hand side is minimized at $\lambda = 4 \frac{t}{A}$, yielding

$$\mathbb{P}[S \geq t] \leq \exp\left(-2 \frac{t^2}{A}\right).$$

Applying the same argument to $-S$ and taking a union bound gives the two-sided inequality.

Proof of Hoeffding without replacement. It suffices to consider an ordered sample $X_1, \dots, X_n$ drawn without replacement from the fixed population $z_1, \dots, z_N$; the unordered subset average has the same distribution. If $n = N$, the sample mean equals the population mean exactly, so assume $n < N$. Let

$$S := \sum_{i=1}^n X_i, \quad \mathcal{F}_k := \sigma(X_1, \dots, X_k), \quad M_k := \mathbb{E}[S \mid \mathcal{F}_k].$$

Then $(M_k)_{k=0}^n$ is a martingale and $M_n - M_0 = S - \mathbb{E}[S]$.

Fix k and condition on $\mathcal{F}_{k-1}$. There are $m = N - k + 1$ population elements still available; write their mean as μ_{k-1} . If the next draw is $X_k = x$, then the remaining $m - 1$ elements have mean

$$\frac{m\mu_{k-1} - x}{m - 1}.$$

Thus the conditional expectation of the final sum changes by

$$M_k - M_{k-1} = \frac{N - n}{N - k} (x - \mu_{k-1}).$$

As x ranges over the remaining population, this martingale difference lies in an interval of length at most

$$\frac{N - n}{N - k} (b - a) \leq b - a.$$

Its conditional mean is zero, so the conditional form of Hoeffding's lemma gives

$$\mathbb{E}[\exp(\lambda(M_k - M_{k-1})) \mid \mathcal{F}_{k-1}] \leq \exp\left(\lambda^2 \frac{(b - a)^2}{8}\right).$$

Iterating over $k = 1, \dots, n$ yields

$$\log(\mathbb{E}[\exp(\lambda(S - \mathbb{E}[S]))]) \leq n\lambda^2 \frac{(b - a)^2}{8}.$$

Chernoff and the same optimization as before give

$$\mathbb{P}[S - \mathbb{E}[S] \geq t] \leq \exp\left(-\frac{2t^2}{n(b - a)^2}\right).$$

Since $\mathbb{E}[S] = n\mu$, set $t = n\varepsilon$ and repeat for the lower tail:

$$\mathbb{P}\left[\left|\frac{1}{n}S - \mu\right| \geq \varepsilon\right] \leq 2 \exp\left(-\frac{2n\varepsilon^2}{(b - a)^2}\right).$$

Proof of McDiarmid's inequality. Let

$$Z := f(X_1, \dots, X_n), \quad \mathcal{F}_i := \sigma(X_1, \dots, X_i), \quad M_i := \mathbb{E}[Z \mid \mathcal{F}_i].$$

Then M_i is the Doob martingale of Z . We have

$$Z - \mathbb{E}[Z] = \sum_{i=1}^n (M_i - M_{i-1}).$$

Define

$$g_{i(x)} := \mathbb{E}[f(X_1, \dots, X_{i-1}, x, X_{i+1}, \dots, X_n) \mid \mathcal{F}_{i-1}].$$

By independence of X_i from $\mathcal{F}_{i-1}$, $M_i = g_{i(X_i)}$ and $M_{i-1} = \mathbb{E}[g_{i(X_i)} \mid \mathcal{F}_{i-1}]$. The bounded-differences hypothesis gives $|g_{i(x)} - g_{i(x')}| \leq c_i$ for all x, x' , so g_i has range at most c_i . Hence the martingale difference $M_i - M_{i-1} = g_{i(X_i)} - \mathbb{E}[g_{i(X_i)} \mid \mathcal{F}_{i-1}]$ has conditional mean zero and conditional range length at most c_i . By conditional Hoeffding's lemma,

$$\mathbb{E}[\exp(\lambda(M_i - M_{i-1})) \mid \mathcal{F}_{i-1}] \leq \exp\left(\lambda^2 \frac{c_i^2}{8}\right).$$

Iterating conditional expectations gives

$$\log(\mathbb{E}[\exp(\lambda(Z - \mathbb{E}[Z]))]) \leq \frac{\lambda^2}{8} \sum_{i=1}^n c_i^2.$$

Chernoff and optimization give

$$\mathbb{P}[Z - \mathbb{E}[Z] \geq t] \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

Apply the same argument to $-f$ for the lower tail.

Proof of Bernstein's inequality. Let $S = \sum_{i=1}^n X_i$ and let $v = \sigma^2 = \sum_i \mathbb{E}[X_i^2]$. For $0 < \lambda < \frac{3}{b}$, using $\mathbb{E}[X_i] = 0$, $|X_i| \leq b$, and the elementary bound $q! \geq 2 \cdot 3^{q-2}$ for $q \geq 2$, we get

$$\begin{aligned} \mathbb{E}[\exp(\lambda X_i)] &= 1 + \sum_{q=2}^{\infty} \lambda^q \frac{\mathbb{E}[X_i^q]}{q!} \leq 1 + \sum_{q=2}^{\infty} \lambda^q \frac{\mathbb{E}[|X_i|^q]}{q!} \\ &\leq 1 + \frac{\lambda^2 \mathbb{E}[X_i^2]}{2} \sum_{q=0}^{\infty} \left(\lambda \frac{b}{3}\right)^q \leq \exp\left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda \frac{b}{3})}\right). \end{aligned}$$

By independence,

$$\log(\mathbb{E}[\exp(\lambda S)]) \leq \frac{\lambda^2 v}{2(1 - \lambda \frac{b}{3})}.$$

Therefore

$$\mathbb{P}[S \geq t] \leq \exp\left(-\lambda t + \frac{\lambda^2 v}{2(1 - \lambda \frac{b}{3})}\right).$$

Choose

$$\lambda = \frac{t}{v + b \frac{t}{3}}.$$

Then $0 < \lambda < \frac{3}{b}$ and

$$-\lambda t + \frac{\lambda^2 v}{2(1 - \lambda \frac{b}{3})} = -\frac{t^2}{2(v + b \frac{t}{3})}.$$

This proves Bernstein's upper-tail inequality; the lower tail follows by applying the same argument to $-X_i$.

All four proofs fit the same template: construct a sum or martingale, prove a conditional MGF bound for each increment, multiply the bounds by independence or the tower rule, and optimize the Chernoff parameter. What changes is the source of the MGF bound: range alone for Hoeffding, finite-population martingale ranges for sampling without replacement, coordinate stability for McDiarmid, and variance plus range for Bernstein.

The payoff: a generalization guarantee for empirical risk minimization

Let $\mathcal{H}$ be a binary hypothesis class and let $Z = (X, Y)$ be drawn from $\mathcal{D}$. For $h \in \mathcal{H}$ define the 0-1 loss

$$\ell_{h(Z)} := 1[h(X) \neq Y],$$

the population risk

$$L_{\mathcal{D}}(h) := \mathbb{E}[\ell_{h(Z)}],$$

and for an i.i.d. sample $S = (Z_1, \dots, Z_n)$ drawn from $\mathcal{D}^n$, the empirical risk

$$L_{n(h)} := \frac{1}{n} \sum_{i=1}^n \ell_{h(Z_i)}.$$

Let

$$\hat{h} \in \operatorname{argmin}_{h \in \mathcal{H}} L_{n(h)}$$

be an empirical risk minimizer.

The goal is to prove that with high probability,

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + O\left(\sqrt{\frac{\log(\Gamma_{\mathcal{H}}(2n)) + \log(\frac{1}{\delta})}{n}}\right).$$

Step 1: reduce ERM to uniform convergence

Define the uniform generalization gap

$$\Delta(S) := \sup_{h \in \mathcal{H}} |L_{\mathcal{D}}(h) - L_{n(h)}|.$$

On any sample S , for every $h \in \mathcal{H}$,

$$L_{\mathcal{D}}(\hat{h}) \leq L_{n(\hat{h})} + \Delta(S) \leq L_{n(h)} + \Delta(S) \leq L_{\mathcal{D}}(h) + 2\Delta(S).$$

Taking the infimum over h gives

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + 2\Delta(S).$$

Thus it remains to bound $\Delta(S)$ with high probability.

Step 2: concentration of the uniform gap around its mean

Changing one example Z_i changes every empirical risk $L_{n(h)}$ by at most $\frac{1}{n}$, and leaves $L_{\mathcal{D}}(h)$ fixed. Therefore $\Delta(S)$ changes by at most $\frac{1}{n}$ when one coordinate of S is changed. McDiarmid's inequality gives, for every $s > 0$,

$$\mathbb{P}[\Delta(S) - \mathbb{E}[\Delta(S)] \geq s] \leq \exp(-2ns^2).$$

Equivalently, with probability at least $1 - \delta$,

$$\Delta(S) \leq \mathbb{E}[\Delta(S)] + \sqrt{\frac{\log(\frac{1}{\delta})}{2n}}.$$

Step 3: symmetrization

Let $S^g = (Z_1^g, \dots, Z_n^g)$ be an independent ghost sample from $\mathcal{D}^n$, and let

$$L_n^{g(h)} := \frac{1}{n} \sum_{i=1}^n \ell_{h(Z_i^g)}.$$

For each fixed h , $\mathbb{E}_{S^g} [L_n^{g(h)}] = L_{\mathcal{D}}(h)$, so $L_{\mathcal{D}}(h) - L_{n(h)} = \mathbb{E}_{S^g} [L_n^{g(h)} - L_{n(h)}]$ pointwise in S . By Jensen's inequality applied to the convex function $|\cdot|$, and then by exchanging sup and $\mathbb{E}_{S^g}$ via monotonicity,

$$|L_{\mathcal{D}}(h) - L_{n(h)}| \leq \mathbb{E}_{S^g} [|L_n^{g(h)} - L_{n(h)}|] \implies \sup_{h \in \mathcal{H}} |L_{\mathcal{D}}(h) - L_{n(h)}| \leq \mathbb{E}_{S^g} \left[\sup_{h \in \mathcal{H}} |L_n^{g(h)} - L_{n(h)}| \right].$$

Taking expectation over S ,

$$\mathbb{E}[\Delta(S)] \leq \mathbb{E}_{S, S^g} \left[\sup_{h \in \mathcal{H}} |L_n^{g(h)} - L_{n(h)}| \right].$$

So the population-vs-empirical problem is reduced to comparing two empirical averages on two independent samples.

Step 4: condition on the combined sample

Condition on the unordered pool of the combined $2n$ observations in $S \cup S^g$. Because the two samples are i.i.d. and exchangeable, after conditioning the only randomness is which n of the $2n$ positions are assigned to S ; this is a uniform split without replacement.

Fix a hypothesis h . On the conditioned pool, the losses $\ell_{h(Z_i)}$ are fixed numbers in $[0, 1]$. Let $L_{P(h)}$ be their average over the full $2n$ -point pool. Since S and S^g are complementary halves,

$$|L_n^{g(h)} - L_{n(h)}| = 2|L_{n(h)} - L_{P(h)}|.$$

Hoeffding without replacement with $N = 2n$ and sample size n gives

$$\mathbb{P} \left[|L_n^{g(h)} - L_{n(h)}| > t \mid S \cup S^g \right] \leq 2 \exp \left(-n \frac{t^2}{2} \right).$$

Now take a union bound over prediction patterns on the conditioned $2n$ feature points. Each loss pattern is determined by a prediction pattern and the fixed labels in the pool, so the number of distinct loss patterns is at most the growth function $\Gamma_{\mathcal{H}}(2n)$. Therefore, with

$$Y := \sup_{h \in \mathcal{H}} |L_n^{g(h)} - L_{n(h)}|,$$

we have

$$\mathbb{P}[Y > t \mid S \cup S^g] \leq 2\Gamma_{\mathcal{H}}(2n) \exp\left(-n\frac{t^2}{2}\right).$$

Since $Y \in [0, 1]$, integrating this tail bound gives

$$\mathbb{E}[Y \mid S \cup S^g] = \int_0^1 \mathbb{P}[Y > t \mid S \cup S^g] dt \leq t_0 + \int_{t_0}^\infty 2\Gamma_{\mathcal{H}}(2n) \exp\left(-n\frac{t^2}{2}\right) dt.$$

Choose

$$t_0 = \sqrt{2\frac{\log(2\Gamma_{\mathcal{H}}(2n))}{n}};$$

the integrand at t_0 equals 1 and the Gaussian tail integrates to at most $\frac{\sqrt{2\pi}}{2}$, so

$$\mathbb{E}[Y \mid S \cup S^g] \leq C\sqrt{\frac{\log(2\Gamma_{\mathcal{H}}(2n))}{n}}$$

for a universal constant C . Taking expectation over the pool,

$$\mathbb{E}[\Delta(S)] \leq C\sqrt{\frac{\log(2\Gamma_{\mathcal{H}}(2n))}{n}}.$$

Step 5: finish the high-probability bound

Combining the expectation bound with McDiarmid's inequality, with probability at least $1 - \delta$,

$$\Delta(S) \leq C\sqrt{\frac{\log(2\Gamma_{\mathcal{H}}(2n))}{n}} + \sqrt{\frac{\log(\frac{1}{\delta})}{2n}}.$$

Absorbing constants and the harmless factor of 2 inside the logarithm,

$$\Delta(S) = O\left(\sqrt{\frac{\log(\Gamma_{\mathcal{H}}(2n)) + \log(\frac{1}{\delta})}{n}}\right).$$

Using the ERM reduction from Step 1,

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + O\left(\sqrt{\frac{\log(\Gamma_{\mathcal{H}}(2n)) + \log(\frac{1}{\delta})}{n}}\right).$$

This is the i.i.d. growth-function ERM guarantee, and it completes the mini-course.

References

- Boucheron, Lugosi, and Massart, *Concentration Inequalities: A Nonasymptotic Theory of Independence*. See Lemma 2.2 for Hoeffding's lemma, Theorem 2.8 for Hoeffding's inequality, Theorem 2.10 and Corollary 2.11 for Bernstein-type bounds, Section 3.2 for the bounded-differences property, and Theorem 6.2 for McDiarmid's inequality.
- Shalev-Shwartz and Ben-David, *Understanding Machine Learning: From Theory to Algorithms*. See Chapters 2–4 for the ERM principle and the agnostic PAC setting, and Chapter 6 for the growth function and the uniform-convergence reduction.

- Vapnik and Chervonenkis, *On the Uniform Convergence of Relative Frequencies of Events to Their Probabilities*, Theory of Probability and Its Applications 16 (1971). This is the original source of the symmetrization argument and the growth-function bound used in the final section.

Homework 3 Submission

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Course: DSC 291 – Learning Theory through Formal Proof and Proof Presentation

Quarter: Spring 2026

Instructor: Tianhao Wang

Part A

Provided in separate file.

Part B

1. **Proof of the No-Free-Lunch theorem.** The universal objects are the learner A , the domain $\mathcal{X}$, and the sample size n with $n < \frac{|\mathcal{X}|}{2}$. The adversary may choose the distribution $\mathcal{D}$ and the realizable target f^* after seeing A and n , but before the random sample is drawn. I write the proof for deterministic A ; for a randomized learner, condition also on its internal random seed, so that the same calculation applies after averaging over that seed.

Proof. Assume $n \geq 1$. For $n = 0$, choose any $x \in \mathcal{X}$, let the target label be the opposite of the fixed prediction $A(())(x)$, and put all mass on that labeled point. Then the learner has loss 1, so the theorem is immediate. Choose a set

$$C = \{x_1, \dots, x_{2n}\} \subseteq \mathcal{X}$$

of $2n$ distinct points, which exist because $2n < |\mathcal{X}|$. For every labeling $f : C \rightarrow \{0, 1\}$, let $\mathcal{D}_f$ be the uniform distribution on the labeled examples

$$\{(x, f(x)) \mid x \in C\}.$$

Extend f arbitrarily to all of $\mathcal{X}$. Then $L_{\mathcal{D}_f}(f) = 0$.

We now average over a uniformly random labeling $F : C \rightarrow \{0, 1\}$, equivalently independent fair bits $F(x)$ for $x \in C$. Draw sample points $X_1, \dots, X_n$ independently and uniformly from C (independently of F), give the learner the labeled sample

$$S = ((X_1, F(X_1)), \dots, (X_n, F(X_n))),$$

and evaluate it under $\mathcal{D}_F$.

Let $U(S_X)$ be the number of points of C that do not appear among $X_1, \dots, X_n$. The output $A(S)$ is a deterministic function of S , hence determined by the unlabeled sample points S_X and the labels observed by the learner. Conditional on these, the label $F(x)$ of any unseen point x is still an independent fair bit, since F is independent of S_X and its coordinates are mutually independent. Hence, for every unseen x , the prediction $A(S)(x)$ is fixed while $F(x)$ is an independent fair bit, so

$$\mathbb{E}[1[A(S)(x) \neq F(x)] \mid S_X, \text{observed labels}] = \frac{1}{2}.$$

Therefore

$$\mathbb{E}_{F,S}[L_{\mathcal{D}_F}(A(S))] \geq \mathbb{E}\left[\frac{U(S_X)}{2n} \cdot \frac{1}{2}\right] = \frac{\mathbb{E}[U(S_X)]}{4n}.$$

For each fixed $x \in C$,

$$\mathbb{P}[x \text{ is unseen}] = \left(1 - \frac{1}{2n}\right)^n \geq \frac{1}{2},$$

where the last inequality follows from Bernoulli's inequality: $\left(1 - \frac{1}{2n}\right)^n \geq 1 - \frac{n}{2n} = \frac{1}{2}$. Thus

$$\mathbb{E}[U(S_X)] = \sum_{x \in C} \mathbb{P}[x \text{ is unseen}] \geq 2n \cdot \frac{1}{2} = n.$$

We have shown

$$\mathbb{E}_{F,S}[L_{\mathcal{D}_F}(A(S))] \geq \frac{1}{4}.$$

Let

$$W := L_{\mathcal{D}_F}(A(S)) \in [0, 1].$$

If $p := \mathbb{P}[W \geq \frac{1}{8}]$, then

$$\mathbb{E}[W] \leq \frac{1}{8} \cdot (1 - p) + 1 \cdot p = \frac{1}{8} + 7\frac{p}{8}.$$

Since $\mathbb{E}[W] \geq \frac{1}{4}$, it follows that $p \geq \frac{1}{7}$. But

$$\mathbb{P}_{F,S}\left[W \geq \frac{1}{8}\right] = \mathbb{E}_F\left[\mathbb{P}_S[L_{\mathcal{D}_F}(A(S)) \geq \frac{1}{8}]\right].$$

Hence there exists a labeling $f^* : C \rightarrow \{0, 1\}$ such that

$$\mathbb{P}_S[L_{\mathcal{D}_{f^*}}(A(S)) \geq \frac{1}{8}] \geq \frac{1}{7}.$$

Taking $\mathcal{D} = \mathcal{D}_{f^*}$ proves the theorem. $\square$

2. **Application to the lower-bound direction of the Fundamental Theorem.** The corollary is: if $\text{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is not PAC learnable, even in the realizable setting.

Proof. Suppose, for contradiction, that a learner A PAC learns $\mathcal{H}$ in the realizable setting. Apply the PAC guarantee with

$$\varepsilon = \frac{1}{9} \quad \text{and} \quad \delta = \frac{1}{8}.$$

Let $n = n_{\mathcal{H}}(\frac{1}{9}, \frac{1}{8})$ be the corresponding sample size. Since $\text{VCdim}(\mathcal{H}) = \infty$, there is a set $C \subseteq \mathcal{X}$ of size $2n$ shattered by $\mathcal{H}$.

Run the averaging argument from the proof of NFL above on this shattered set C . That argument uses only that C has $2n$ points; the hypothesis $n < \frac{|\mathcal{X}|}{2}$ in the theorem statement served only to guarantee such a set exists, which here is supplied directly by infinite VC dimension. It produces a distribution $\mathcal{D}$ supported on C and a labeling f^* of C such that

$$\mathbb{P}\left[L_{\mathcal{D}}(A(S)) \geq \frac{1}{8}\right] \geq \frac{1}{7}.$$

Because C is shattered, some $h^* \in \mathcal{H}$ agrees with f^* on all of C . Since $\mathcal{D}$ is supported on C , we have $L_{\mathcal{D}}(h^*) = 0$. Thus $\mathcal{D}$ is realizable by $\mathcal{H}$.

But the PAC guarantee would require

$$\mathbb{P}\left[L_{\mathcal{D}}(A(S)) > \frac{1}{9}\right] \leq \frac{1}{8},$$

while NFL gives, using $\frac{1}{8} > \frac{1}{9}$,

$$\mathbb{P}\left[L_{\mathcal{D}}(A(S)) > \frac{1}{9}\right] \geq \mathbb{P}\left[L_{\mathcal{D}}(A(S)) \geq \frac{1}{8}\right] \geq \frac{1}{7} > \frac{1}{8}.$$

This is a contradiction. □

A concrete infinite-VC class is the class of finite-subset indicators on $\mathbb{N}$:

$$\mathcal{H}_{\text{fin}} := \{h_A : \mathbb{N} \rightarrow \{0, 1\} \mid A \subseteq \mathbb{N} \text{ finite}\}, \quad h_{A(x)} := 1[x \in A].$$

This class has infinite VC dimension. Indeed, for any finite set $S \subseteq \mathbb{N}$ and any labeling of S , let $T \subseteq S$ be the positively labeled points. Since T is finite, $h_T \in \mathcal{H}_{\text{fin}}$ realizes exactly that labeling on S . Thus every finite S is shattered. Applying the corollary, $\mathcal{H}_{\text{fin}}$ is not PAC learnable.

3. **Worked construction.** Let the domain be

$$\mathcal{X} = \{a, b, c\}$$

and consider the following learning rule for one training example: after seeing $S = ((x_1, y_1))$, output

$$A(S)(x) := \begin{cases} y_1 & \text{if } x = x_1 \\ 0 & \text{otherwise.} \end{cases}$$

This is the memorization rule that predicts 0 away from the observed point. The domain has three points so that the single training example ($n = 1$) satisfies the NFL hypothesis $n < \frac{|\mathcal{X}|}{2}$; the bad distribution constructed below is supported on the two-point set $\{a, b\}$.

Define the target

$$f^*(a) = 1, \quad f^*(b) = 1, \quad f^*(c) = 0,$$

and let $\mathcal{D}$ be uniform on $(a, 1)$ and $(b, 1)$. Then $L_{\mathcal{D}}(f^*) = 0$. If $S = ((a, 1))$, the learned rule is wrong on b and correct on a , so

$$L_{\mathcal{D}}(A(S)) = \frac{1}{2}.$$

If $S = ((b, 1))$, the learned rule is wrong on a and correct on b , so again

$$L_{\mathcal{D}}(A(S)) = \frac{1}{2}.$$

Therefore

$$\mathbb{P}_{S \text{ drawn from } \mathcal{D}}\left[L_{\mathcal{D}}(A(S)) \geq \frac{1}{8}\right] = 1.$$

This example makes the NFL mechanism explicit: the learner has no information about an unobserved point, and the adversary puts the opposite of the learner's default behavior there.

4. **Comparison with the Week 1 No-Free-Lunch theorem.** The Week 1 theorem lives in an online prediction setting on a finite domain. The learner predicts labels sequentially, and the proof adversary is adaptive: it can choose the next label after seeing the learner's prediction. Because the learner is deterministic, the entire interaction is predetermined, so these adaptive choices can be folded into a single fixed target f and ordering – which is why the statement can be phrased without reference to

an adaptive adversary. The conclusion is deterministic. For a deterministic learner, there exist a target function and an ordering of the domain on which the learner makes a mistake on every round.

The Week 3 theorem lives in the batch i.i.d. / PAC setting. The adversary is oblivious to the realized sample: it chooses a distribution and target labeling before sampling occurs. The conclusion is probabilistic. With constant probability over the random training sample, the learner's output has constant population loss, even though the distribution is realizable.

The two statements are quantitatively different because they measure different failures. Week 1 measures the number of online mistakes against an adaptive sequence. Week 3 measures population risk after a random training sample. They are both called “No-Free-Lunch” because the underlying obstruction is the same: without structural restrictions on the target class, labels on unseen points are unconstrained, so no learner can be uniformly reliable.

Part C

1. I used AI (in a human loop) to write and improve the mini course and the solution to part B.
2. I accepted the general proofs/solutions/sketch it provided.
3. I don't like the writing style of the AI, so I read through the whole solutions and constantly ask it to change phrasing and adapt the write-up, or I change it myself.
4. I read through everything by myself and didn't find any mistakes. (I also asked AI to check it carefully again in the end.)

AI Workflow. Since Codex is faster and allows for more usage in \$20 plan, but Claude can have a more detailed look sometimes, I found it helpful to use Codex for “initial generation” and Claude for improving the manuscript. I also added more instructions to the `CLAUDE.md`/`AGENTS.md` for my typesetting standards (like using $\mathbb{E}[\dots]$ for expectation and $\mathbb{P}[\dots]$ for probability).